WORKED SOLUTION

STEP I 2005, Question 12

Probability and Statistics • Venn Diagram Bounds

STEP I 2005, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/05-s1-q12

The key idea

Questions of the form "what range of values is consistent?" are about a Venn diagram whose regions are unknown but must all be *non-negative*. Write every region in terms of as few free unknowns as possible; each condition "this region ≥ 0" then becomes a bound, and the bounds together give the range. In part (ii), a counting argument (how many times each wizard is counted when we add the three single probabilities) gets the "exactly two" probability without needing the individual regions.

Solution

Part (i)

Let S be the event that a hobbit smokes a pipe and H the event that it wears a hat. Put $t = \mathrm{P}(S \cap H)$. The four regions of the Venn diagram are

$$\mathrm{P}(S \cap H) = t, \qquad \mathrm{P}(S \cap H') = 0.7 - t,$$
$$\mathrm{P}(S' \cap H) = 0.4 - t, \qquad \mathrm{P}(S' \cap H') = 1 - (0.7 + 0.4 - t) = t - 0.1.$$

Any choice of t that makes all four non-negative gives a genuine probability distribution, and conversely. The binding conditions are $t \geq 0.1$ (from the last region) and $t \leq 0.4$ (from the third); the others then follow. So $0.1 \leq t \leq 0.4$, and since $p = 0.7 - t$,

$$0.3 \leq p \leq 0.6.$$

Both ends are attained: $p = 0.6$ when as few pipe-smokers as possible wear hats (every hobbit does at least one of the two), and $p = 0.3$ when every hat-wearer smokes.

Part (ii): exactly two

Let H, C, R be the events that a wizard wears a hat, a cloak and a ring. Let a, b be the probabilities that a wizard wears exactly one and exactly two of the three items. We

know the probability of exactly none is 0.05 and of exactly three is 0.1, so

$$0.05 + a + b + 0.1 = 1 \quad \Longrightarrow \quad a + b = 0.85.$$

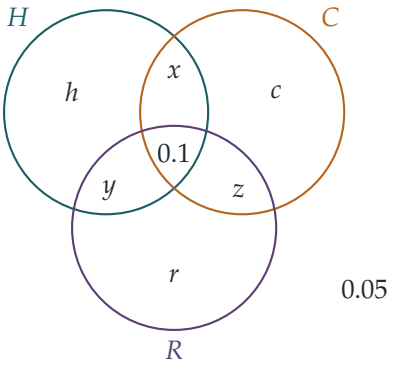
Now add the three given probabilities: $\mathrm{P}(H) + \mathrm{P}(C) + \mathrm{P}(R) = 0.7 + 0.8 + 0.4 = 1.9$. In this sum a wizard who wears exactly k items is counted k times, so

$$a + 2b + 3(0.1) = 1.9 \quad \Longrightarrow \quad a + 2b = 1.6.$$

Subtracting, $b = 0.75$ (and $a = 0.1$). The probability that a wizard wears exactly two of the items is $\frac{3}{4}$.

Part (ii): the range of q

Label the seven inside regions as in the diagram: h, c, r for exactly one item, x, y, z for exactly two, with 0.1 in the centre and 0.05 outside.



From the totals for H, C, R, and from part (ii) so far,

$$h + x + y = 0.6, \qquad c + x + z = 0.7,$$
$$r + y + z = 0.3, \qquad x + y + z = 0.75.$$

Using the last equation to eliminate a pair from each of the first three,

$$h = z - 0.15, \qquad c = y - 0.05, \qquad r = x - 0.45.$$

(As a check, $h + c + r = 0.75 - 0.65 = 0.1 = a$.) So the whole distribution is determined by x, y, z with $x + y + z = 0.75$, and it is a genuine distribution exactly when all six regions are non-negative:

$$x \geq 0.45, \qquad y \geq 0.05, \qquad z \geq 0.15.$$

Ask yourself How large can x be?

Since $x = 0.75 - y - z$ and $y + z \geq 0.05 + 0.15 = 0.2$, we get $x \leq 0.55$, with equality when $y = 0.05$, $z = 0.15$. The smallest value $x = 0.45$ is attained with, for example,

$y = 0.1$, $z = 0.2$. Every value in between is attained too, since we can move y and z continuously. Hence

$$0.45 \le x \le 0.55.$$

Finally, wearing a hat but not a ring while wearing a cloak is exactly the region x, so

$$q = \mathrm{P}(H \cap R' \mid C) = \frac{\mathrm{P}(H \cap C \cap R')}{\mathrm{P}(C)} = \frac{x}{0.8}, \qquad \text{giving} \qquad \frac{9}{16} \le q \le \frac{11}{16}.$$

Checks

- At the extreme $x = 0.55$, $y = 0.05$, $z = 0.15$ we get $h = 0$, $c = 0$, $r = 0.1$; the eight regions sum to 1 and give $\mathrm{P}(H) = 0.7$, $\mathrm{P}(C) = 0.8$, $\mathrm{P}(R) = 0.4$, as required.
- We enumerated every assignment of the eight regions in steps of 0.01 consistent with the data: every one has $x + y + z = 0.75$, and q runs from 0.5625 to 0.6875. Similarly for (i), p runs from 0.3 to 0.6.

Common slips

- In (i), forgetting the region "neither" and so getting the lower bound $t \ge 0$ instead of $t \ge 0.1$, which gives the wrong upper limit $p \le 0.7$.
- In (ii), dividing by 1 instead of by $\mathrm{P}(C) = 0.8$: the condition "given that he wears a cloak" changes the denominator.
- Treating H, C, R as independent. Nothing in the question says they are, and in fact $\mathrm{P}(H \cap C \cap R) = 0.1 \neq 0.7 \times 0.8 \times 0.4$.

Beyond STEP

Finding the range of a quantity that is linear in unknown probabilities, subject to linear equations and non-negativity, is a small *linear programme*: the extremes always occur at corners of the feasible region, which is why ours came from setting regions to zero. The same idea gives the Fréchet bounds $\max(0, \mathrm{P}(A) + \mathrm{P}(B) - 1) \le \mathrm{P}(A \cap B) \le \min(\mathrm{P}(A), \mathrm{P}(B))$ used in part (i). For more on inclusion–exclusion and conditional probability, read Blitzstein and Hwang, *Introduction to Probability*.

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